Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
1   #include<stdio.h>
2   void func1(int *arr){
3     for(int i=1; i<=3;i++)
4       (i-1)[arr] = (i%3);
5   }
6   void func2 (int a[1] , int arr){
7     a[0] = arr + 1;
8   }
9  int main(){
10   int arr[4] = {0,0,0,0};
11   func1(arr+1);
12   for(int i=0; i<4; i++)
13           printf("%d ", i[arr] );
14   printf("\n");
15   func2(arr, *arr);
16   for(int i=0; i<4; i++)
17           printf("%d ", i[arr] );
18   return 0;
19 }
```

a[1] = arr[0]
a[0]
a[1] → 0 1 2 0
0 1 2 0
1

Output
0120
1120

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
  int temp = arr + 1;
  a = &temp;
}
```

Explain your answer

yes, since the address is being passed

Q. 2: Write the output of the following sequence of instructions **long arr2[100]; int arr1[100], a1=(&arr2[50]), a2=(&arr2[10]), b1=(&arr1[50]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

8
10 11
a 8
60 68
50 50
8x50
50x60
~~50x8~~
40x8
40x4

320 160

Since int size is 4 bytes and array is a consecutive block of element we do (50x4 − 10x4) and for long size is 8 bytes so (50x8 − 10x8)

50x4 − 10x4
4(40) = 160